

## Department Overview

- **Department Name:** Department of Civil Engineering
- **Faculty:** Faculty of Engineering & Architecture
- **Accreditation & Framework:** Fully accredited by the **Pakistan Engineering Council (PEC)**. The department implements **Outcome Based Education (OBE)** and **Outcome-Based Assessment (OBA)** models at the undergraduate level as recommended by the Engineering Accreditation Board (EAB) of the PEC.
- **History:** Actively contributing to educational infrastructure and academia since 2008. The postgraduate (MS) program was introduced in 2015.
- **Research Focus Note:** Highlights an active engagement in **Affordable Housing Research** aligning with national construction organizations aimed at providing housing facilities to homeless populations and boosting economic activity in Pakistan.

## Mission

To produce competent professionals of high ethical values, enabling them for executive positions in civil engineering, higher studies in prestigious universities, life-long learning, and societal leadership. Also, to provide a dynamic learning environment that emphasizes sustainable engineering practices, problem-solving skills, teamwork, communication, and promotes cutting-edge research and leadership skills.

## Program Educational Objectives (PEOs)

Civil engineering graduates are expected to demonstrate the following attributes 3 to 5 years post-graduation:

1. **PEO-1:** Essential knowledge and adequate civil engineering skills to meet industrial requirements.
2. **PEO-2:** High ethical values and societal responsibilities for sustainable development.
3. **PEO-3:** Effective communication, strong interpersonal skills, and the capacity to lead and work collaboratively in teams.
4. **PEO-4:** Ability to keep abreast with the latest innovations and technological developments through continued professional practices, higher studies, and research.

## 1. Bachelor of Science in Civil Engineering (BS Civil)

### Admission Requirements

- **Academic Background:** F.Sc (Pre-Engineering) from a recognized board or an equivalent certification with at least **60% marks**, OR a Diploma of Associate Engineer (DAE) in the relevant field with a minimum of **60% marks**.
- **Testing:** GAT General or the university's internal NTS test.

### Degree Requirements

- **Total Credit Hours:** 136

- **Total Courses:** 46
- **Minimum Academic Standing:** CGPA  $\geq$  2.0

## Curriculum Plan (Program Schema by Semester)

*Note: Credit hour listings follow the format (Theory + Lab = Total).*

| Semester | Course Code   | Course Title                                 | Prerequisite     | Credit Hours  |
|----------|---------------|----------------------------------------------|------------------|---------------|
| 1st      | CIVILE-206    | Engineering Materials                        | None             | 2 + 1 = 3     |
|          | ENGG-201      | Engineering Drawing                          | None             | 1 + 2 = 3     |
|          | HUM-161       | Reading & Writing Skills                     | None             | 2 + 0 = 2     |
|          | MATHa-117     | Applied Calculus                             | None             | 3 + 0 = 3     |
|          | HUM-102 / 101 | Pakistan Studies / Islamic Studies or Ethics | None / None      | 1+0=1 / 2+0=2 |
| 2nd      | CIVILE-207    | Engineering Surveying-I                      | None             | 2 + 1 = 3     |
|          | GEOE-302      | Engineering Geology                          | None             | 2 + 1 = 3     |
|          | ENGG-205      | Engineering Mechanics                        | None             | 3 + 1 = 4     |
|          | MATHA-214     | Differential Equations                       | Applied Calculus | 3 + 0 = 3     |

|            |                     |                                                  |                              |                 |
|------------|---------------------|--------------------------------------------------|------------------------------|-----------------|
|            | ENGG-204            | Basic Electro-Mechanical Engineering             | None                         | $2 + 1 = 3$     |
|            | CIVILE-303          | Civil Engineering Drawing & Graphics             | Engineering Drawing          | $1 + 2 = 3$     |
| <b>3rd</b> | CS-114              | Programming Fundamentals                         | None                         | $2 + 1 = 3$     |
|            | CIVILE-314          | Engineering Surveying-II                         | Engineering Surveying-I      | $2 + 1 = 3$     |
|            | CIVILE-302          | Mechanics of Solids-I                            | Engineering Mechanics        | $2 + 1 = 3$     |
|            | Econ-105 / ENGG-206 | Engineering Economics / Fluid Mechanics-I        | None / None                  | $2+0=2 / 3+1=4$ |
|            | CIVILE-304          | Quantity & Cost Estimation                       | None                         | $2 + 1 = 3$     |
| <b>4th</b> | CIVILE-419 / 208    | Construction Engineering / Structural Analysis-I | None / Engineering Mechanics | $3+0=3 / 3+0=3$ |
|            | CIVILE-306          | Soil Mechanics                                   | None                         | $3 + 1 = 4$     |
|            | MATHA-216           | Numerical Analysis                               | None                         | $3 + 0 = 3$     |

|            |                       |                                                            |                         |                   |
|------------|-----------------------|------------------------------------------------------------|-------------------------|-------------------|
|            | ENGG-321              | Environmental Engineering-I                                | None                    | $2 + 1 = 3$       |
|            | CIVILE-208            | Concrete Technology                                        | None                    | $1 + 1 = 2$       |
| <b>5th</b> | MATHA-217             | Probability and Statistics                                 | None                    | $2 + 1 = 3$       |
|            | ENGG-209              | Fluid Mechanics-II                                         | Fluid Mechanics-I       | $3 + 1 = 4$       |
|            | HUM-268 /<br>MGMT-304 | Communication &<br>Presentation Skills / Org.<br>Behaviour | None / None             | $2+0=2$ / $2+0=2$ |
|            | CIVILE-309 /<br>310   | Reinforced Concrete Design-I /<br>Transportation Planning  | None / None             | $3+0=3$ / $3+0=3$ |
| <b>6th</b> | ENGG-441              | Construction Management                                    | None                    | $2 + 1 = 3$       |
|            | CIVILE-418            | Mechanics of Solids-II                                     | Mechanics of Solids-I   | $2 + 1 = 3$       |
|            | CIVILE-403            | Highway and Traffic Engineering                            | Transportation Planning | $3 + 1 = 4$       |
|            | CIVILE-402 /<br>315   | Engineering Hydrology /                                    | None / Structural       | $2+1=3$ / $3+0=3$ |

|            |                  |                                                 |                              |               |
|------------|------------------|-------------------------------------------------|------------------------------|---------------|
|            |                  | Structural Analysis-II                          | Analysis-I                   |               |
|            | HUM-276          | Technical Report Writing                        | None                         | 0 + 1 = 1     |
| <b>7th</b> | MGMT-315 / 311   | Entrepreneurship / Architecture & Town Planning | None / None                  | 3+0=3 / 3+0=3 |
|            | CIVILE-404       | Geotechnical and Foundation Engineering         | Soil Mechanics               | 3 + 1 = 4     |
|            | CIVILE-407       | Reinforced Concrete Design–II                   | Reinforced Concrete Design-I | 3 + 1 = 4     |
|            | CIVILE-406L      | Civil Engineering Project-I                     | None                         | 0 + 3 = 3     |
| <b>8th</b> | CIVILE-401       | Geoinformatics                                  | Engineering Surveying-II     | 1 + 1 = 2     |
|            | CIVILE-412 / 411 | Environmental Engineering-II / Steel Structures | Environmental Eng.-I / None  | 2+0=2 / 3+0=3 |
|            | CIVILE-410       | Hydraulics and Irrigation Engineering           | Fluid Mechanics-II           | 3 + 1 = 4     |

|  |             |                              |      |             |
|--|-------------|------------------------------|------|-------------|
|  | HUM-112     | Professional Ethics          | None | $2 + 0 = 2$ |
|  | CIVILE-417L | Civil Engineering Project-II | None | $0 + 3 = 3$ |

## 2. Master of Science in Civil Engineering (MS Civil)

### Admission Requirements

- **Academic Background:** 16 years of education or an equivalent benchmark qualification (e.g., a 4-year B.E. or BS degree) in Civil Engineering from an HEC-recognized institution.
- **Minimum Marks:** At least **60% marks** under the annual grading setup, or a minimum program CGPA of **2.5 out of 4.0**.
- **Testing:** A valid **GAT General** result with a passing score of at least **50% marks**.

### Degree Requirements

- **Total Credit Hours:** 30
- **Total Coursework Requirements:** 8 advanced postgraduate courses.
- **Minimum Academic Standing:** CGPA  $\geq$  2.5